

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

Objective type Questions

1. Choose the correct alternative for each of the
following questions : $2 \times 10 = 20$

(a) Number of states of FSM required
simulated behaviour of a computer with
amemory capable of storing m words

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(1)

(Turn over)

each of length n :

(i) $m \times 2^n$

~~(ii)~~ 2^{mn}

~~(iii)~~ $2^{(m+n)}$

(iv) All of these

(b) Which of the following pairs of regular expression are equivalent ?

(i) $1(01)^*$ and $(10)^*1$

(ii) $X(XX)^*$ and $(XX)^*X$

(iii) X^* and X^*X^{*-}

(iv) All of these

(c) Given a NFA with N states, the maximum number of states in an equivalent minimize DFA is at least :

(i) N^2

~~(ii)~~ 2^N

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(2)

Contd.

(iii) $2N$

~~(iv)~~ $N!$

(d) The set of all strings over the alphabet

$S = \{a, b\}$ (including ϵ) is denoted by :

~~(i)~~ $(a+b)^*$

(ii) $(a+b)^+$

(iii) a^*b^*

(iv) a^*b^*

(e) The logic of pumping lemma is a good example of :

~~(i)~~ Pigeon hole principle

(ii) Divide and conquer technique

(iii) Recursion

(iv) Iteration

(f) A language is regular if and only if :

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(3)

(Turn over)

- (i) Accepted by DFA
 - (ii) Accepted by PDA
 - (iii) Accepted by LBA
 - (iv) Accepted by Turing Machine
- (g) A regular grammar is :
- (i) Context free grammar
 - (ii) Non-context free grammar
 - (iii) English grammar
 - (iv) None of these
- (h) Regular expressions are closed under :
- (i) Union
 - (ii) Intersection
 - (iii) Kleen star
 - (iv) All of these

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(4)

Contd.

- (i) There are ____ tuples in finite state machine :
- (i) 4
 - (ii) 5
 - (iii) 6
 - (iv) Unlimited
- (j) Transition function maps :
- (i) $\Sigma^*Q \rightarrow \Sigma$
 - (ii) $Q^*Q \rightarrow \Sigma$
 - (iii) $\Sigma^*\Sigma \rightarrow Q$
 - (iv) $Q^*\Sigma \rightarrow Q$

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

$$5 \times 4 = 20$$

(5)

(Turn over)

2. Construct a NFA to accepts exactly those strings that have the symbol 1 in the second last position.

3. Define regular expression. Explain the application of regular expression.

4. Construct DFA for a regular expression $a^*b(a+b)^*$.

5. State and prove the pumping lemma for regular language $L = \{a^n b^n \mid n \geq 1\}$, $n=1,2,\dots$ is not a regular language.

6. What is difference between DFA and NFA ?

7. What is ambiguity grammar ? Explain with example.

8. Construct push down automate that accept $L = \{a^n b^n \mid n \geq 1\}$.

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(6)

Contd.

9. Obtained a context free grammar of the language having $L = \{a^n b^n \mid n > 0\}$.

Section-C

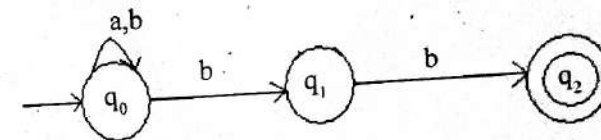
Long Answer type Questions

Answer any two questions of the following :

15×2=30

10. Define non-deterministic finite automata.

Convert the following NFA to equivalent deterministic finite automata.



11. What is push down automata ? Construct push down automata for the language $L = \{a^n b^n \mid n \geq 1\}$.

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(7)

(Turn over)

Q2. What is Turing machine ? Design a Turing machine of the language $L = \{a^n b^{2n} | n \geq 1\}$.

Q3. Define DFA. Minimize the following DFA.

